Cognitive Modeling

Complete Lecture Notes

Week 12: Gibbs Sampling and Convergence Diagnostics

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Chapter 1

Parameter Estimation with MCMC

1.1 Week 12: Gibbs Sampling and Convergence Diagnostics

> Chapter 5: Parameter estimation with data-driven methods (MCMC) · Lecture 1 h · Lab 2 h Running data: log response times of one participant (a normal model with unknown mean and precision), and the power forgetting model sampled with last week's Metropolis sampler.

1.1.1 Key terms

Important terms of this lesson, with their Persian equivalents (repeated even if they appeared in earlier weeks).

English	فارسی	توضیح کوتاه
Gibbs sampling	نمونه‌گیری Gibbs	به‌روزرسانی هر پارامتر از توزیع شرطی کاملش
convergence	همگرایی	رسیدن زنجیره‌ها به توزیع هدف
parameter	پارامتر	کمیت قابل تنظیم مدل که بین افراد یا شرایط فرق می‌کند
estimate / estimator	برآورد / برآوردگر	روش به / مقدار به‌دست‌آمده برای پارامتر دست آوردن آن
Markov chain Monte Carlo (MCMC)	زنجیره‌ی مارکوف Monte Carlo	نمونه‌گیری از پسین با یک زنجیره‌ی تصادفی
response time (RT)	زمان واکنش	فاصله‌ی ارائه‌ی محرک تا پاسخ
forgetting curve	منحنی فراموشی	کاهش یادآوری با گذشت زمان
Metropolis algorithm	الگوریتم Metropolis	پیشنهاد یک حرکت و پذیرش یا رد آن

English	فارسی	توضیح کوتاه
sampler	نمونه‌گیر (sampler)	الگوریتمی که از توزیع نمونه می‌گیرد
full conditional	توزیع شرطی کامل (full conditional)	توزیع یک پارامتر به شرط بقیه و داده
conjugate prior	پیشین مزدوج (conjugate)	پیشینی که پسین هم‌خانواده می‌دهد
R-hat	R-hat	مقایسه‌ی واریانس بین و درون زنجیره‌ها؛ باید ≈ 1 باشد
effective sample size (ESS)	اندازه‌ی مؤثر نمونه (ESS)	تعداد نمونه‌ی مستقل هم‌ارز
chain	زنجیره	دنباله‌ی نمونه‌های MCMC
trace plot	نمودار trace	مقدار نمونه‌ها در برابر شماره‌ی تکرار
Monte Carlo	Monte Carlo (مونت‌کارلو)	تقریب با میانگین نمونه‌های تصادفی
proposal	پیشنهاد (proposal)	مقدار جدیدی که الگوریتم امتحان می‌کند
posterior	توزیع پسین (posterior)	باور درباره‌ی پارامتر بعد از دیدن داده
MCSE	خطای استاندارد Monte Carlo	خطای شبیه‌سازی در میانگین پسین
prior	توزیع پیشین (prior)	باور درباره‌ی پارامتر پیش از دیدن داده
HMC / NUTS	Hamiltonian Monte Carlo / NUTS	MCMC مبتنی بر گرادیان
burn-in / warm-up	دوره‌ی گرم‌شدن (burn-in)	نمونه‌های اولیه که دور ریخته می‌شوند
autocorrelation	خودهمبستگی	وابستگی نمونه‌های پشت سر هم
sensitivity (d′)	حساسیت	فاصله‌ی توزیع علامت و نویز

1.1.2 Learning outcomes

By the end of this week, students will be able to:

1. Derive the full conditional distributions of a semi-conjugate normal model and implement a Gibbs sampler from scratch.
2. Explain when Gibbs sampling is efficient and when it mixes slowly (correlated parameters), and how Metropolis-within-Gibbs handles non-conjugate conditionals.
3. Implement split-$\hat{R}$ and the effective sample size (ESS), and verify them against ArviZ.
4. Diagnose non-convergence from multiple chains using trace plots, $\hat{R}$, ESS and the Monte Carlo standard error.
5. Apply a reporting checklist for MCMC results.

1.1.3 Motivation & big picture

Last week's Metropolis sampler had one tuning knob, the proposal SD, and a nagging question: *how do we know the chain has converged?* This week answers both concerns.

Gibbs sampling removes the tuning problem for a large and important class of models. Instead of proposing a jump for all parameters and sometimes rejecting it, it updates one parameter at a time, drawing it *exactly* from its distribution given the current values of all the others. Every draw is accepted. Many classic Bayesian programs (BUGS, JAGS, used throughout Lee & Wagenmakers and Kruschke) are built on this idea.

Convergence diagnostics are the honest part of MCMC. A chain is only useful if it has (a) forgotten its starting point and (b) explored the whole posterior, and nothing in a single chain proves either. The standard remedy is to run several chains from different starting points and check that they agree. $\hat{R}$ quantifies "do the chains agree?", and the ESS quantifies "how much independent information do the correlated draws contain?". No MCMC result should be reported without them.

Where is this used?

- Every published MCMC analysis must report convergence diagnostics; reviewers in psychology and neuroscience now routinely ask for $\hat{R}$ and ESS.
- Gibbs sampling powers the classic Bayesian programs BUGS and JAGS, which are used in many cognitive modeling textbooks (Lee & Wagenmakers, Kruschke) and in ecology, epidemiology and education research.
- Image processing and communications: Gibbs samplers have been used for image restoration and for detection in multi-antenna (MIMO) receivers.
- Topic models (finding themes in large text collections) were classically fitted with Gibbs sampling.

1.1.4 Lecture notes

In plain words Metropolis needs a well-chosen step size. Gibbs sampling avoids that for many models by updating one parameter at a time: fix all the other parameters at their current values, and draw the chosen parameter directly from its exact distribution under those conditions (its full conditional). Then move to the next parameter, and so on. Nothing is ever rejected. The second half of the week answers the uncomfortable question "has my sampler actually worked?". We run several chains from very different starting points. If they all end up describing the same distribution, that is good evidence of convergence; $\hat{R}$ ("R-hat") is a single number that measures whether the chains agree (it should be below 1.01). Because neighboring draws are correlated, 4,000 draws may contain only the information of, say, 1,000 independent ones: that number is the effective sample size (ESS), and it tells us how precise our averages are.

Suggested timing (60 min): Gibbs sampling 20 · when Gibbs struggles 5 · convergence and multiple chains 8 · $\hat{R}$ 10 · ESS and MCSE 10 · reporting checklist 7.

1. Gibbs sampling

For parameters $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_k)$, the full conditional of θ_m is its posterior given the data and *all other* parameters, $p(\theta_m \mid \boldsymbol{\theta}_{-m}, \mathrm{y})$. Gibbs sampling cycles through them:

1. Start at $\boldsymbol{\theta}^{(0)}$.
2. For $s = 1, \ldots, S$:
   - draw $\theta_1^{(s)} \sim p(\theta_1 \mid \theta_2^{(s-1)}, \ldots, \theta_k^{(s-1)}, \mathrm{y})$
   - draw $\theta_2^{(s)} \sim p(\theta_2 \mid \theta_1^{(s)}, \theta_3^{(s-1)}, \ldots, \mathrm{y})$
   - ... (always using the newest values)
   - draw $\theta_k^{(s)} \sim p(\theta_k \mid \theta_1^{(s)}, \ldots, \theta_{k-1}^{(s)}, \mathrm{y})$

It is a special case of Metropolis–Hastings in which the proposal is the full conditional itself, so the acceptance probability is always 1.

Finding full conditionals. Write the joint density $p(\mathrm{y} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})$, keep only the factors that contain θ_m, and recognize a known distribution. This works when priors are (conditionally) conjugate.

Example: a normal model for log RTs. Let y_i be log response times, $y_i \sim \mathcal{N}(\mu, 1/\sqrt{\tau})$, where $\tau = 1/\sigma^2$ is the precision. Priors: $\mu \sim \mathcal{N}(m_0, s_0)$ and $\tau \sim \mathrm{Gamma}(a_0, \text{rate } b_0)$, independent. Then:

$$\mu \mid \tau, \mathrm{y} \sim \mathcal{N}\left(\frac{m_0/s_0^2 + \tau \sum_i y_i}{1/s_0^2 + N\tau},\ \ \frac{1}{\sqrt{1/s_0^2 + N\tau}} \right),$$

$$\tau \mid \mu, \mathrm{y} \sim \mathrm{Gamma}\left(a_0 + \frac{N}{2},\ \ \text{rate} = b_0 + \frac{1}{2}\sum_i (y_i - \mu)^2 \right).$$

Reading the formulas. The conditional mean of μ is a precision-weighted average of the prior mean and the data mean (as in the beta-binomial of Week 9). The conditional distribution of τ adds half an observation to the shape and half a squared residual to the rate for each data point.

> Note on parameterizations. NumPy's `rng.gamma(shape, scale)` uses scale = 1/rate. Mixing up rate and scale is the most common bug in Gibbs samplers.

Reading the math

- $\boldsymbol{\theta}_{-m}$: "all parameters except θ_m".
- Precision $\tau = 1/\sigma^2$: the inverse of the variance. Large precision = small noise. Working with precisions makes combining information a simple addition.
- The μ update: the conditional mean $\dfrac{m_0/s_0^2 + \tau \sum y_i}{1/s_0^2 + N\tau}$ is a weighted average of the prior mean m_0 (weight $1/s_0^2$, the prior's precision) and the data mean (total weight $N\tau$, the data's precision). The conditional precision is the *sum* of the two precisions. Worked example: $1{+}4{\times}10 = 41$, so the data dominate.

- The τ update: a Gamma distribution whose shape grows by $1/2$ per observation and whose rate grows by half of each squared residual. Its mean, shape/rate, is roughly "1 / (average squared residual)", i.e. the precision suggested by the data.

2. When Gibbs struggles, and Metropolis-within-Gibbs

- Strong posterior correlations. Gibbs moves parallel to the coordinate axes. If the posterior is a narrow diagonal ridge (like the A–d trade-off of the power model), each step can move only a tiny distance, and the chain zig-zags slowly along the ridge. Remedies: reparameterize, update correlated parameters jointly ("blocking"), or use gradient-based samplers (Week 13).
- Non-conjugate conditionals. Most cognitive models (forgetting curves, SDT, the GCM) have no closed-form conditionals. Metropolis-within-Gibbs updates each parameter with a one-dimensional Metropolis step targeting its full conditional, and uses exact Gibbs draws where available.

> Advanced: JAGS and BUGS automatically pick a sampler for each parameter (conjugate Gibbs, slice sampling, adaptive Metropolis), which is why they run on almost any model. Modern Hamiltonian Monte Carlo (Stan, PyMC's NUTS) instead uses the gradient of the log posterior to move all parameters jointly along the posterior's geometry, and handles correlated posteriors far better (Week 13).

3. Convergence: the logic of multiple chains

An MCMC chain has converged when its draws are representative of the posterior. Two failure modes: 1. Not yet stationary: the chain still reflects the starting value. 2. Stuck or slow: the chain explores only part of the posterior (e.g., one of two modes, or one end of a long ridge).

Neither can be detected reliably from one chain, because a chain stuck in one region can look perfectly stable. Run $C \geq 4$ chains from over-dispersed starting points: if they all end up describing the same distribution, that is good (though never conclusive) evidence of convergence.

Trace plots of all chains overlaid should look like one well-mixed "hairy caterpillar", with no chain visibly separated or trending.

4. The potential scale reduction factor $\hat{R}$

Compare the variance between chains with the variance within chains. With C chains of length S (after burn-in), chain means $\bar{\theta}_c$ and within-chain variances s_c^2:

$$W = \frac{1}{C}\sum_c s_c^2, \qquad B = \frac{S}{C-1}\sum_c (\bar{\theta}_c - \bar{\theta})^2, \qquad \widehat{\text{var}}^+ = \frac{S-1}{S}W + \frac{1}{S}B, \qquad \hat{R} = \sqrt{\frac{\widehat{\text{var}}^+}{W}}.$$

If the chains have not mixed, B is large relative to W and $\hat{R} > 1$. As chains converge, $\hat{R} \to 1$.

- Split-$\hat{R}$: split each chain into a first and a second half and treat them as $2C$ chains. This also detects a single chain that is still drifting.
- Threshold: the classic rule was $\hat{R} < 1.1$; current recommendations (Vehtari et al., 2021) use $\hat{R} < 1.01$, computed after rank-normalization (as ArviZ does).

5. Effective sample size and Monte Carlo standard error

Autocorrelated draws carry less information than independent ones. The effective sample size is

$$\text{ESS} = \frac{C \cdot S}{1 + 2\sum_{h=1}^{\infty} \rho_h},$$

where ρ_h is the lag-h autocorrelation, estimated from all chains and truncated when the estimates become noise. ESS is the number of *independent* draws that would give the same precision.

- The Monte Carlo standard error of a posterior mean is MCSE $=$ posterior SD$/\sqrt{\text{ESS}}$. It must be small relative to the precision you report (e.g., don't report 3 decimals if the MCSE is 0.01).
- Rule of thumb: bulk-ESS $\geq$ 400 (with 4 chains) for reliable posterior means and intervals; tail-ESS $\geq$ 400 for reliable extreme quantiles.

Reading the math

- $\hat{R}$: W is the average variance *inside* each chain; B measures how far the chain *means* are from each other. $\widehat{\text{var}}^{+}$ is an estimate of the total posterior variance that includes the disagreement between chains. If the chains agree, B adds almost nothing and $\hat{R} = \sqrt{\widehat{\text{var}}^{+}/W} \approx 1$. In the "not converged" worked example, the two chains sit around 1.0 and 2.0, B is huge and $\hat{R} = 4.4$.
- ESS: $C \cdot S$ is the total number of draws; the denominator $1 + 2\sum \rho_h$ inflates for correlated draws. With $\rho_h = 0.5^h$, the sum is $0.5 + 0.25 + \cdots = 1$, so the draws count only one third: 4,000 draws are worth 1,333 independent ones.
- MCSE $=$ posterior SD$/\sqrt{\text{ESS}}$ is the simulation error of the reported mean; decide the number of decimals from it.

6. MCMC reporting checklist

1. Sampler, number of chains, iterations, burn-in (warm-up), and any thinning.
2. Starting values (dispersed?).
3. $\hat{R}$ (which version) and ESS for all reported parameters, plus the maximum $\hat{R}$ and minimum ESS.
4. Trace plots (in a supplement) and a statement that they were inspected.
5. Priors, and the posterior summaries with MCSE-appropriate precision.
6. A posterior predictive check (Week 10).

For an engineer

- Precision-weighted averaging in the μ update is the same rule as maximal-ratio combining or the Kalman measurement update: combine estimates in proportion to their inverse noise variances.
- Gibbs sampling updates one variable given all the others, like coordinate-wise iterative algorithms (alternating optimization) and the iterative exchange of conditional information in turbo-style receivers; Gibbs samplers have been proposed for MIMO detection.
- $\hat{R}$ is a one-way ANOVA (between-group vs. within-group variance) across independent simulation runs: the same check you would do by running a Monte Carlo simulation with several seeds and verifying that the runs agree.
- ESS $= N/(1+2\sum \rho_h)$ is the standard formula for the effective number of independent samples of a correlated random process: averaging N correlated samples reduces the variance only as much as averaging ESS independent ones (related to the correlation time).

For the instructor

- Draw Gibbs on a contour plot. Two parameters, an elliptical posterior. Gibbs moves in horizontal and vertical steps only (a staircase). Then draw a narrow diagonal ellipse: the staircase steps become tiny. That one picture explains both how Gibbs works and when it struggles.
- Make $\hat{R}$ visual first: overlay four trace plots, once with chains that agree and once with one chain stuck elsewhere. Ask "do these chains describe the same distribution?" before showing any formula.
- Teach the reporting checklist as a habit. Ask students to apply it to their project and to every MCMC result in the lab.
- Typical questions.
  - *"$\hat{R}$ is 1.00, so my model is correct?"* No. $\hat{R}$ only checks that the chains agree; the model can still be wrong (check with a PPC) or the code can still be buggy.
  - *"Should I thin my chains?"* Not for correctness; it only saves memory.
  - *"Rate or scale?"* NumPy's Gamma uses scale = 1/rate; the most common bug in student Gibbs samplers.

1.1.5 Worked example

Part 1: one Gibbs iteration by hand

Data: $N = 4$ log RTs, y $= (6.0, 6.3, 6.1, 6.4)$, so $\sum y_i = 24.8$. Priors: $m_0 = 6$, $s_0 = 1$, $a_0 = 2$, $b_0 = 0.1$. Current values: $\mu = 6.0$, $\tau = 10$.

Update μ. Precision $= 1/1^2 + 4 \times 10 = 41$; mean $= (6/1 + 10 \times 24.8)/41 = (6 + 248)/41 = 6.195$; SD $= 1/\sqrt{41} = 0.156$. Draw, say, $\mu = 6.22$.

Update τ given $\mu = 6.22$. Squared residuals: $0.0484 + 0.0064 + 0.0144 + 0.0324 = 0.1016$. Shape $= 2 + 4/2 = 4$; rate $= 0.1 + 0.1016/2 = 0.1508$. The conditional mean of τ is $4/0.1508 = 26.5$ (i.e., $\sigma \approx 0.19$). Draw, say, $\tau = 24.1$.

New state: $(\mu, \tau) = (6.22, 24.1)$. No accept/reject step was needed.

Part 2: $\hat{R}$ by hand

Two chains of $S = 4$ draws each.

Not converged. Chain 1: (1.0, 1.2, 0.8, 1.0); chain 2: (2.0, 2.2, 1.8, 2.0). - Chain means: 1.0 and 2.0. Within-chain variances (ddof = 1): both $0.08/3 = 0.0267$, so $W = 0.0267$. - $B = S \times \text{var}(\text{means}) = 4 \times 0.5 = 2.0$. - $\widehat{\text{var}}^{+} = \frac{3}{4} \times 0.0267 + \frac{1}{4} \times 2.0 = 0.520$. - $\hat{R} = \sqrt{0.520/0.0267} = 4.4$. The chains clearly disagree.

Converged (realistic numbers). $S = 1000$ draws per chain, $W = 0.05$, chain means 1.00 and 1.02: $B = 1000 \times \text{var}(1.00, 1.02) = 1000 \times 0.0002 = 0.2$; $\widehat{\text{var}}^{+} = 0.999 \times 0.05 + 0.2/1000 = 0.05015$; $\hat{R} = \sqrt{0.05015/0.05} = 1.0015$. Below 1.01: fine.

Part 3: ESS by hand

$C \cdot S = 4000$ draws with autocorrelations $\rho_h = 0.5^h$. Then $\sum_{h \geq 1} \rho_h = 1$, so ESS $= 4000/(1 + 2) = 1333$. With a posterior SD of 0.2, MCSE $= 0.2/\sqrt{1333} = 0.0055$: report the posterior mean to two decimals.

1.1.6 Lab (2 hours)

Goals

1. Implement a Gibbs sampler for the normal model of log RTs and validate it against a grid posterior.
2. Run four dispersed chains and inspect them.
3. Implement split-$\hat{R}$ and ESS; compare with ArviZ.
4. Break convergence on purpose (the power model with a bad Metropolis sampler) and watch the diagnostics catch it.

Timing: Part A 30 min · Part B 15 min · Part C 30 min · Part D 30 min · Part E 15 min.

Setup

Code · کد: `code/week12.py`, cell 1 · سلول 1

Part A: a Gibbs sampler (30 min)

> Code · کد: `code/week12.py`, cell 2 · سلول 2

Expected output:

```
N = 40, sample mean = 6.200, sample SD = 0.225
Gibbs: mu = 6.1994 (SD 0.0358), sigma = 0.2283 (SD 0.0255)
Grid:  mu = 6.1996, sigma = 0.2287
```

Part B: four dispersed chains (15 min)

> Code · کد: `code/week12.py`, cell 3 · سلول 3

Expected output:

```
Chain means of mu: [6.2006 6.1988 6.1996 6.1992]
```

Gibbs forgets even absurd starting values within a few iterations here, because μ and τ are nearly uncorrelated in this posterior.

Part C: $\hat{R}$ and ESS from scratch, checked against ArviZ (30 min)

> Code · کد: `code/week12.py`, cell 4 · سلول 4

Expected output:

```
       R-hat (ours)  R-hat (ArviZ)  ESS (ours)  bulk-ESS (ArviZ)  MCSE
       ↪  of mean
param
mu           1.0002         1.0004   9980.4260        10032.6243
↪  0.0004
sigma        0.9997         0.9998   8772.9605         8861.7940
↪  0.0003
      ess_bulk ess_tail r_hat mcse_mean  mcse_sd
mu       10032     9427  1.00   0.00036  0.00027
sigma     8861     9714  1.00   0.00027  0.00021
```

(The exact column layout of `az.summary` differs slightly between ArviZ versions; the output above is from ArviZ 1.x.) Our simple estimators agree closely with ArviZ. With $4 \times 2500 = 10\,000$

draws, ESS ≈ 9000–10 000 means the Gibbs draws are almost independent. ArviZ additionally rank-normalizes the draws (Vehtari et al., 2021), which makes its $\hat{R}$ and bulk-ESS robust to heavy tails, so small differences are expected. In practice use ArviZ; today's point is to know what it computes.

Part D: catching non-convergence (30 min)

We reuse last week's Metropolis sampler on the power forgetting model, once badly tuned and short, once well tuned.

Code · کد: `code/week12.py`, cell 5 · سلول 5

Expected output:

```
bad (tiny steps, short)   split-R-hat(d) = 3.583, ArviZ R-hat(d) =
↪  3.424, ESS(d) = 4, chain means of d = [0.248 0.785 0.451 0.328]
good (tuned, longer)      split-R-hat(d) = 1.002, ArviZ R-hat(d) =
↪  1.002, ESS(d) = 1025, chain means of d = [0.414 0.414 0.415 0.413]
```

What to notice: with tiny steps and short chains, each chain is still close to where it started. The traces are separate, $\hat{R}$ is far above 1.01 and the ESS is tiny. *Each chain on its own looks smooth and stable*, which is exactly why one chain is never enough. The tuned sampler gives overlapping traces, $\hat{R} = 1.002$ and an ESS of about 1000 from 20 000 draws. The Gibbs sampler of Part C, with almost uncorrelated draws, reached an ESS of about 9000–10 000 from 10 000 draws.

Part E: interpret and report (15 min)

Write an MCMC methods-and-results paragraph for the log-RT model, following the checklist in Lecture §6: sampler, chains, iterations, burn-in, starting values, $\hat{R}$, ESS, posterior means and 95% intervals of μ and σ (back-transform μ to milliseconds with e^{μ} and say what it represents).

Exercises (increasing difficulty)

Exercise 1 ⋆. Show by algebra that the full conditional of τ is a Gamma distribution: write $p(\tau \mid \mu, \mathrm{y}) \propto \tau^{N/2} \exp(-\frac{\tau}{2}\sum(y_i - \mu)^2) \cdot \tau^{a_0-1} e^{-b_0\tau}$ and read off the shape and rate.

Exercise 2 ⋆. Introduce the rate/scale bug (`scale=B0+...` instead of `1/(...)`) into `gibbs_normal`. Do the diagnostics catch it? Does the validation against the grid? What does this tell you about the difference between *convergence* and *correctness*?

Exercise 3 ⋆⋆ (correlated Gibbs). Write a Gibbs sampler for a bivariate normal "posterior" with correlation ρ (full conditionals: $\theta_1 \mid \theta_2 \sim \mathcal{N}(\rho\theta_2, \sqrt{1-\rho^2})$ and vice versa). Compare the ESS for $\rho = 0, 0.9, 0.99$ with 5000 iterations, and plot the zig-zag path for $\rho = 0.99$.

Exercise 4 ⋆⋆ (Metropolis-within-Gibbs). Sample the power forgetting model by updating A and d one at a time, each with a 1-D Metropolis step. Compare the ESS per second with the joint Metropolis sampler. Why might it be worse here? (Hint: the A–d correlation.)

Exercise 5 ⋆⋆⋆ (Advanced: two conditions). Extend the Gibbs sampler to log RTs from two conditions with separate means μ_1, μ_2 and a shared precision τ. Simulate data with a true difference of 0.1, sample, and report the posterior of $\mu_2 - \mu_1$ and $P(\mu_2 > \mu_1 \mid \text{y})$ with diagnostics.

Lab check-off

Show: (1) the Gibbs-vs-grid validation (Part A), (2) your $\hat{R}$/ESS table next to ArviZ's (Part C), and (3) the bad-vs-good trace plots with their diagnostics (Part D).

1.1.7 Connection to AI & LLMs

Gibbs sampling from a masked language model. Models like BERT are trained to fill in a masked word given all other words: they learn full conditionals, $P(w_i \mid \text{all other words})$. That is exactly what a Gibbs sampler needs. Repeatedly masking one position and resampling it generates whole sentences (Wang & Cho, 2019). The toy model below knows only the two full conditionals of a two-word "language", and Gibbs sampling recovers the joint distribution of sentences. Because the language is "sticky" (dog goes with barks, cat with meows), the chain mixes slowly, and $\hat{R}$ catches this in short runs.

Code · کد: `code/week12.py`, cell 6 · سلول 6

Expected output:

```
  50 iterations: P(subject = dog) = 0.285 (exact 0.35), R-hat = 1.151
3000 iterations: P(subject = dog) = 0.354 (exact 0.35), R-hat = 1.000
```

Try it (no API needed). - ⋆ Make the language even stickier (e.g., 0.33 on the diagonal and 0.001 elsewhere, renormalized). How many iterations do you now need for $\hat{R} < 1.01$? Compute the ESS (Week 12, Part C) of the `is_dog` chains. - ⋆⋆ Real masked models are trained separately for each position, so their conditionals need not come from any single joint distribution. Perturb `p_v_given_s` slightly (without changing `p_s_given_v`) and check whether the Gibbs sampler still recovers `joint`. What does this tell you about trusting a sampler whose conditionals are learned?

Optional reading: Wang, A., & Cho, K. (2019). BERT has a mouth, and it must speak: BERT as a Markov random field language model. *arXiv:1902.04094*.

1.1.8 Common mistakes & misconceptions

1. Rate vs. scale in the Gamma distribution. NumPy and SciPy use scale = 1/rate. Write down which one your formula uses.
2. Using old values within a Gibbs sweep. Each conditional must use the *newest* values of the other parameters.
3. "$\hat{R} \approx 1$, so the results are correct." Diagnostics test agreement between chains, not correctness of the model or the code (Exercise 2). Validate new samplers on known cases.
4. One long chain instead of several. A stuck chain can look perfectly stable. Use ≥ 4 dispersed chains.
5. Reporting results with low ESS. With ESS = 50, a posterior mean has an MCSE of about 14% of the posterior SD, and tail quantiles are unreliable.
6. Thinning to "fix" autocorrelation. Thinning lowers storage but does not increase the information. ESS already accounts for autocorrelation.
7. Using the 1.1 threshold uncritically. Modern guidance is $\hat{R} < 1.01$ with rank-normalized split-$\hat{R}$.
8. Diagnosing only some parameters. Check *all* reported parameters, including derived ones.

1.1.9 Check your understanding

1. Why does Gibbs sampling never reject a draw? Answer Each draw comes exactly from the full conditional distribution. Viewed as Metropolis–Hastings with the full conditional as proposal, the acceptance ratio is always 1.

2. What do you need to be able to run a pure Gibbs sampler? Answer Full conditional distributions for every parameter (or block) that you can sample from directly, typically because the priors are conditionally conjugate.

3. A Gibbs sampler for two strongly correlated parameters has a very low ESS. Why, and what could you do? Answer Gibbs updates one coordinate at a time, so on a narrow diagonal ridge every step is tiny and the chain zig-zags slowly. Remedies: reparameterize to reduce the correlation, update the correlated parameters jointly (blocking), or use HMC/NUTS.

4. Explain $\hat{R}$ in one sentence. Which value indicates convergence? Answer It compares the variance between chains with the variance within chains (it estimates how much the posterior spread could shrink if the chains ran forever); values near 1 (< 1.01 by current guidance) indicate that the chains agree.

5. You have 4000 draws and an ESS of 200 for a parameter with posterior SD 0.5. What is the MCSE of its posterior mean? How many decimals should you report? Answer MCSE $= 0.5/\sqrt{200} = 0.035$. Report the mean to one decimal (or two at most), because the third decimal is Monte Carlo noise.

6. Why can't one chain demonstrate convergence? Answer A chain stuck in one region (one mode, or one part of a ridge) can look stationary and well mixed, and nothing in the chain itself reveals the unexplored regions. Only chains started elsewhere, which end up somewhere different, expose the problem.

7. Name four things a methods section should report about an MCMC analysis. Answer Any four of: sampler/software; number of chains; iterations and warm-up; starting values; $\hat{R}$ and ESS for all parameters; that trace plots were inspected; priors; a posterior predictive check.

1.1.10 Assignment

A5: MCMC from scratch, with diagnostics (1.0 point of 20; released Week 12, due at the start of the Week 14 lecture)

Data: the payoff SDT experiment from Week 4, Part D (regenerate it with the same seed: `np.random.default_rng(11)`), and the log-RT data from today's lab (`np.random.default_rng(12)`).

Tasks.

1. Metropolis for SDT (3 pts). Implement a random-walk Metropolis sampler for the shared-d' payoff model (d', c_1, c_2, c_3) with priors $d' \sim \mathcal{N}(1, 1.5)$ and $c_m \sim \mathcal{N}(0, 1)$. Tune the proposal (report the acceptance rate), and run 4 dispersed chains.
2. Diagnostics (2 pts). Report split-$\hat{R}$ and ESS for all parameters with your own functions *and* with ArviZ; show trace plots. State whether you would trust the results and why.
3. Inference (1.5 pts). Report posterior means and 95% intervals; compute the posterior of $c_{\text{conservative}} - c_{\text{liberal}}$ and the probability that it is positive. Compare d' with the Week 4 MLE.
4. Gibbs (2 pts). Implement the two-condition Gibbs sampler of Exercise 5 and report the posterior of the mean difference with diagnostics.
5. Interpretation (1.5 pts, max. 300 words). What do the payoffs do to sensitivity and bias, according to the posterior? Compare with the likelihood-ratio conclusion of Week 4.

Rubric: points as above; within each task: correctness 60%, figures/tables 20%, interpretation 20%. Late submissions lose 10% per day (max. 3 days).

1.1.11 Further reading

- Kruschke (2014), chapter 7 (sections on Gibbs sampling and on MCMC representativeness, accuracy and efficiency): diagnostics, ESS and MCSE explained visually. *(Main reading.)*
- Martin, Kumar & Lao (2021), chapter 2. Exploratory analysis of Bayesian models: trace plots, $\hat{R}$, ESS and MCSE with ArviZ.
- Lee & Wagenmakers (2014), chapter 4. Inferences with Gaussians: the normal model whose Gibbs sampler you built today, in its Bayesian-modeling context.

- *Optional, outside the course list:* Vehtari, A., Gelman, A., Simpson, D., Carpenter, B., & Bürkner, P.-C. (2021). Rank-normalization, folding, and localization: An improved $\hat{R}$ for assessing convergence of MCMC. *Bayesian Analysis, 16*(2), 667–718.

Preview for Week 13: writing samplers by hand teaches how they work, but it is slow and error-prone. Next week we describe models in PyMC, let its NUTS sampler do the work, and use the time we save for the most useful model class in cognitive science: hierarchical models of many participants.